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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers

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Abstract— Bridging communication between the deaf and hearing communities demands reliable automated sign language interpretation. Widespread unfamiliarity with sign language among the general population creates substantial accessibility challenges. This paper presents SignAI, a video-based deep learning system for recognizing Indian Sign Language (ISL). The architecture centers on VideoMAE, a vision transformer pre-trained through masked autoencoding on video data, which simultaneously encodes hand shape and motion cues from gesture clips.

Unlike prior methods that depend on CNNs or LSTM-based recurrent networks, SignAI treats each gesture as a continuous temporal sequence. The pipeline processes raw video into 16-frame clips, applies them to a fine-tuned VideoMAE backbone, and maps the extracted representations to one of 101 ISL categories. End-to-end deployment pairs a React-based upload interface with a FastAPI inference server that returns both text and synthesized speech.

Experiments on the ISL-CSLTR benchmark show a Top-1 validation accuracy of approximately 74%, with well-balanced precision, recall, and F1-score, confirming the suitability of transformer-based sequence models for capturing the fine-grained spatiotemporal structure of ISL gestures.

Keywords— Indian Sign Language (ISL), Gesture Recognition, VideoMAE, Vision Transformer, Deep Learning, Spatiotemporal Analysis, Sequence-Based Classification, Computer Vision

I. INTRODUCTION

For individuals who are deaf or hard of hearing, sign language constitutes the primary medium of everyday communication, weaving together hand configurations, facial expressions, and body orientation to form a complete visual-spatial language. Yet this language remains largely inaccessible to the broader population, creating a persistent social barrier. Automated gesture recognition systems address this gap by translating sign sequences into text or spoken words that non-signers can understand.

Recent strides in computer vision and neural network design have accelerated progress in this field. Early approaches depended on hand-engineered features and classical classifiers, which could not adequately represent the dual spatial-temporal nature of gestures. The subsequent adoption of convolutional-recurrent architectures partially solved the temporal modeling challenge, but these pipelines typically struggle with long-range dependencies spanning many frames.

The emergence of transformer networks marked a turning point. Their self-attention mechanism allows each token in a sequence to attend to every other token, capturing global context that convolutional or recurrent stages miss. Extending this idea to video, Video Masked Autoencoders (VideoMAE) pre-train a transformer encoder by reconstructing heavily masked video patches, yielding rich spatiotemporal representations that transfer well to downstream recognition tasks.

SignAI builds on VideoMAE to tackle ISL recognition. The ISL-CSLTR dataset, comprising 7,671 videos spanning 101 gesture categories, provides the training and evaluation corpus. Videos are uniformly sampled to 16-frame sequences, normalized, and fed to the fine-tuned model. A full-stack deployment with a React frontend and FastAPI backend completes the system, returning prediction text and optional audio to end users.

The system attains roughly 74% Top-1 accuracy, demonstrating robustness across diverse signers and recording conditions while acknowledging residual confusion among visually similar gestures.

Main Contributions

- VideoMAE is applied to ISL video data, enabling joint encoding of spatial hand configurations and temporal motion dynamics within a single transformer pass.
- A reproducible preprocessing pipeline handles frame extraction, uniform sampling to $N=16$ frames, resizing to 224×224 , and ImageNet-based normalization.
- Sequence-level inference replaces per-frame classification, allowing the model to reason about gesture trajectory and continuity rather than isolated snapshots.
- Self-attention across the full clip enables the network to identify which spatial regions and time steps carry the most discriminative gesture information.

- A production-ready pipeline connects a React video-upload frontend to a FastAPI backend, outputting recognized gestures as text with optional text-to-speech.
- Comprehensive evaluation employs accuracy, precision, recall, F1-score, and AUC-ROC to characterize model performance across all 101 gesture classes.

II. LITERATURE SURVEY

A. Deep Learning for Sign Language Understanding

The last decade has seen machine learning displace hand-crafted approaches in gesture recognition. Rastgoo et al. [1] surveyed this trajectory comprehensively, tracing the shift from feature engineering toward end-to-end neural pipelines covering both static hand poses and dynamic gesture sequences. Among early neural architectures, Koller et al. [2] demonstrated that coupling CNN-derived spatial descriptors with hidden Markov model sequence decoders could substantially improve continuous recognition accuracy. Huang et al. [3] took a complementary route, eliminating explicit temporal boundary detection by training directly on unsegmented video streams.

B. Attention and Transformer Models in Gesture Recognition

Transformer architectures introduced a fundamentally different approach to sequence modelling that quickly found application in vision tasks. Camgoz et al. [4] pioneered a fully attention-based pipeline for simultaneous sign recognition and translation, reporting gains in capturing cross-frame dependencies that recurrent networks frequently miss. Their results motivated the broader adoption of attention mechanisms whenever long-range temporal reasoning is required.

C. Convolutional Networks for Gesture Classification

Convolutional networks formed the workhorse of gesture classification for several years. Kumar et al. [8] applied CNN classifiers to ISL hand shapes and achieved strong results within the finite vocabulary they studied. Ko et al. [15] and Pigou et al. [9] similarly leveraged convolutional feature extraction on both still images and short video clips; however, all these methods share a structural limitation: convolution over individual frames cannot capture gestural dynamics that unfold over time.

D. Spatiotemporal Modelling

To overcome the temporal blindspot of standard CNNs, researchers designed architectures that jointly process spatial appearance and temporal motion. Shi et al. [10] and Zhou et al. [11] independently proposed networks combining 2D spatial filters with temporal modules, showing that explicitly encoding motion cues improves continuous recognition benchmarks. These spatiotemporal models set an important precedent for treating video as more than a collection of independent images.

E. Detection-Oriented and Hybrid Architectures

A separate research thread has explored integrating object detectors with gesture classifiers. Tamura et al. [14] coupled YOLO-based hand localization with downstream classification to achieve high frame-level detection accuracy. Aly and Aly [6] demonstrated that unified deep learning pipelines could handle both static poses and dynamic gestures, reflecting the architectural flexibility afforded by modern networks.

F. Recent Work on Complex and Two-Handed Signs

As datasets grew richer, research focus shifted to more challenging scenarios involving compound motions and bimanual coordination. Li et al. [12] and Li et al. [13] separately proposed frameworks for dynamic two-handed ISL recognition, illustrating how deep architectures are evolving to handle the full expressive range of sign languages rather than simplified single-hand subsets.

G. Research Gap and Motivation

A clear gap persists in the literature: most methods either process frames independently, rely on sequential encoders that struggle with long-range context, or focus on hand detection without modelling full gesture sequences. Environmental variation, intra-class gesture variability, and the need for large labelled datasets compound these difficulties.

Detection-first architectures such as YOLO excel at pinpointing hands in individual frames but do not encode the temporal flow that distinguishes one ISL phrase from another. CNN-LSTM hybrids improve on this but still face gradient-decay issues over longer temporal windows. The present work responds to these shortcomings by adopting VideoMAE, which pre-trains on large-scale video via masked reconstruction and can thus attend to extended gesture trajectories without the sequential bottlenecks of recurrent models.

III. METHODOLOGY

SignAI processes ISL gestures as ordered temporal sequences rather than individual frames, capturing the motion continuity that distinguishes sign language from static pose recognition. The workflow proceeds through five stages: frame extraction, preprocessing, sequence formation, spatiotemporal feature learning with VideoMAE, and softmax classification over 101 gesture categories. A confidence gate filters low-certainty predictions before the output is rendered as text and optional speech.

3.1 Dataset

Training and evaluation rely on the ISL-CSLTR benchmark [footnote: ¹ ISL-CSLTR: Indian Sign Language Dataset for Continuous Sign Language Translation and Recognition, Mendeley Data, accessed 2 December 2024], which contains 7,671 video clips and 144,448 individual frames distributed across 101 ISL phrase categories. Clips record different signers performing sentence-level gestures under varying illumination, clutter, and execution speeds. This diversity is intentional—it encourages models that generalize beyond the recording conditions of any single session. The corpus is partitioned into training, validation, and test splits to allow unbiased performance estimation.

3.2 Frame Extraction

Each input video $V = \{f_1, f_2, \dots, f_n\}$ is decoded into its constituent frames. To impose a uniform clip length regardless of original video duration, $N=16$ frames are drawn via uniform temporal sampling. This strategy preserves the gesture's essential motion profile whether the recording is brief or extended, and keeps the sequence length fixed for batch processing.

3.3 Preprocessing

Every sampled frame is independently resized to 224×224 pixels, matching the spatial resolution expected by the VideoMAE encoder. Pixel intensities are then normalized using ImageNet channel statistics:

$$I_{norm} = (I - \mu) / \sigma$$

where μ and σ are the per-channel mean and standard deviation computed over ImageNet. This calibration mitigates the effect of lighting differences across clips and accelerates gradient-based optimization during fine-tuning.

3.4 Sequence Formation

The preprocessed 16 frames are concatenated into a single sequence $S = \{f_1, f_2, \dots, f_{16}\}$. For videos longer than the target window, a sliding-window strategy generates multiple overlapping sequences, augmenting the training distribution. Each sequence constitutes one complete gesture instance passed to the feature extractor.

3.5 Patch Embedding

The VideoMAE encoder begins by partitioning every frame into non-overlapping 16×16 patches. A 224×224 frame⁸ yields 196 patches per frame. Each patch is flattened and linearly projected into a learnable embedding vector, after which learned positional encodings are added so that the transformer can reconstruct both where (spatial) and when (temporal) each patch occurred within the clip.

3.6 Spatiotemporal Feature Extraction via VideoMAE

The sequence of patch embeddings enters the VideoMAE transformer stack. Multi-head self-attention at each layer relates every patch to every other patch across both the spatial and temporal axes:

$$\text{Attention}(Q, K, V) = \text{Softmax}(QK^T / \sqrt{d_k}) V$$

Here Q, K, V are query, key, and value projections of the input, and d_k is the key dimension used for scale normalisation. This formulation lets the network dynamically weight informative spatial regions—e.g., moving hand positions—and informative time steps—e.g., the onset and offset of a gesture—simultaneously, a capability that neither purely spatial CNNs nor unidirectional RNNs provide.

3.7 Classification

A linear classification head maps the transformer's pooled output to logits z_i for each of the $C=101$ gesture classes. A softmax layer converts these logits to a probability distribution:⁹

$$P(y_i) = e^{z_i} / \sum_j e^{z_j}$$

The predicted class is the argmax of this distribution. A confidence threshold θ is applied post-softmax; predictions where $\max P(y_i) < \theta$ are suppressed to reduce erroneous outputs on ambiguous inputs.

3.8 System Integration

The deployed application uses a React web interface for video upload and result display, and a FastAPI server that hosts the inference pipeline. Uploaded video is decoded server-side, processed through the full pipeline, and the resulting gesture label is returned to the client as text. An optional text-to-speech module converts the label to audio, broadening accessibility for users who benefit from spoken output.

Fig. 3.1: Overall workflow of the proposed SignAI system

Fig. 3.6.1: Architecture of the VideoMAE Model for Spatiotemporal Feature Extraction

IV. RESULTS AND DISCUSSION

2 SignAI was assessed on the full ISL-CSLTR corpus: 7,671 video clips, 144,448 frames, and 101 gesture categories. Evaluation employed accuracy, precision, recall, F1-score, and AUC-ROC to give a multi-faceted view of classification quality and generalization.

The fine-tuned VideoMAE backbone achieved a Top-1 validation accuracy of 74%, with precision at 77% and both recall and F1-score at 74%. These closely matched figures indicate that the model avoids the precision-recall trade-offs that can inflate one metric at the expense of another, and that spatiotemporal representations extracted from 16-frame sequences are well-suited to sentence-level ISL categorization.

Training dynamics reveal that accuracy climbs sharply in the first five epochs before levelling off near 74% on the validation set, while the training loss descends monotonically. A modest gap between training and validation loss is present from epoch 7 onward, consistent with mild overfitting on sequence data—a well-known behaviour in video transformers trained on relatively small corpora. The gap remains contained, however, indicating acceptable generalization to unseen signers and conditions.

The confusion matrix displays a dominant diagonal, confirming that the great majority of gesture classes are predicted correctly. Off-diagonal errors concentrate on pairs of gestures sharing similar hand configurations or partial motion trajectories, which is expected given the inherent ambiguity of visually proximate signs. The precision-recall curve yields a macro-average precision of approximately 0.79, and the ROC analysis returns a macro-average AUC of 0.98, signifying excellent class separability across all 101 categories.

Fig. 4.1: Training and validation accuracy over epochs

Fig. 4.2: Training and validation loss over epochs

Fig. 4.3: 13 Confusion matrix for ISL gesture classification

Fig. 4.4: Precision–Recall curve showing model performance

Fig. 4.5: ROC curve for multi-class classification

A. 17 Comparison with Baseline Approaches

Table 4.1 contrasts SignAI against CNN, CNN-LSTM, and YOLO-based baselines. 1 The CNN model scores 65% accuracy; the CNN-LSTM hybrid improves to 69% by incorporating sequential context. The YOLO detector reaches the highest raw accuracy (85%) but operates frame-by-frame and therefore cannot model the temporal flow of continuous gestures. VideoMAE records 74% accuracy alongside the highest AUC-ROC of 0.98, reflecting superior probabilistic calibration and class discrimination.

Table 4.1: Performance Comparison of Different Approaches

Model	Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC
CNN-Based	Spatial	65	67	64	65	0.85
CNN + LSTM	Hybrid	69	70	68	69	0.88
YOLO-Based	Detection	85	87.5	85	85	0.90
VideoMAE	Transformer	74	77	74	74	0.98

Although YOLO records the best per-frame accuracy, this metric conflates localization with recognition and does not reward temporal coherence. The VideoMAE architecture’s lower raw accuracy is accompanied by far stronger AUC-ROC performance, underscoring its ability to produce well-calibrated predictions across all 101 gesture classes rather than excelling on a few dominant ones. For sentence-level ISL interpretation, temporal coherence is non-negotiable; hence VideoMAE is the more appropriate architecture for deployment.

V. CONCLUSION

This paper introduced SignAI, a full-stack system for automated Indian Sign Language recognition built around the VideoMAE vision transformer. By treating each gesture as a fixed-length video sequence rather than a series of independent frames, the architecture captures the continuous hand motion and temporal context essential for reliable ISL interpretation. The accompanying React-FastAPI pipeline makes the system accessible through a standard browser interface, returning recognized gestures as text and optionally as synthesized speech.

Evaluated on the ISL-CSLTR dataset across 101 gesture categories, SignAI achieved 74% Top-1 validation accuracy with a macro-average AUC-ROC of 0.98—metrics that reflect both accurate per-class prediction and excellent probability calibration. While detection-centric baselines such as YOLO attain higher frame-level accuracy, they lack the temporal reasoning needed for continuous gesture streams, making them unsuitable for real-world ISL deployment.

Remaining challenges include confusion among visually similar signs and sensitivity to extreme lighting or background variation. Future directions include data augmentation strategies targeting rare gesture classes, lightweight model distillation for edge deployment, and expansion to additional ISL vocabulary to support broader communicative contexts. SignAI represents a meaningful step toward intelligent assistive technology that fosters equal communication access for deaf and hard-of-hearing communities.

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